

Daily Work Report – ARBE Module

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1. Stock Price, Volume, & Technical Signals (Intraday + Daily)

These APIs give:

- Live price
- % change
- Volume
- OHLC
- Breakouts
- Indicators (RSI, MACD, etc.)

API	Cost	Speed	Accuracy	Rate Limits	Ease	Notes
Yahoo Finance (RapidAPI)	Paid (\$0–\$10/mo)	Fast	Very good	Excellent	Easiest	Best for India
Alpha Vantage	Free / Paid	Slow (free)	Good	Low (5/min)	Medium	Good technical indicators
Twelve Data	Free / Paid	Fast	Good	Medium	Easy	Good for realtime
Finnhub	Paid	Very Fast	Excellent	High	Easy	Costly
Tiingo	Paid	Fast	Very good	High	Medium	Good accuracy

2. Market Index Signals (NIFTY, BANKNIFTY, SENSEX, GLOBAL INDICES)

Needed for:

- NIFTY movement
- Sector rotation
- Market trend signals

API	Cost	Accuracy	Issues
NSE API	Free	Very accurate	Hard (cookies, headers, blocking)
Yahoo Finance	Free/Paid	Very good	Minor delays (1–2 sec)
Alpha Vantage	Free	Medium	Not great India index coverage

3. Sector Performance & Sector Rotation Signals

Sector-level data is harder to get.

API	Ease	Speed	Notes
NSE Sector Indices	Hard	Fast	Needs scraping, often blocked
Yahoo Finance Sector ETFs	Easy	Fast	Very stable

4. Mutual Fund Signals

Needed for:

- NAV movement
- Fund type
- Fund performance trends

API	Accuracy	Ease	Notes
AMFI Official	100% accurate	Medium	Raw text file, needs parse
Value Research	Good	Easy	Not fully official

5. Macroeconomic Indicators

Used for macro updates:

- Inflation
- Interest rates
- GDP
- INR USD
- Crude Oil
- Gold
- VIX

Source	What it gives	Ease	Cost
TradingEconomics API	Inflation, GDP, rates	Easy	Paid
Yahoo Finance Commodities	Oil, gold, USD/INR	Easy	Free/Paid
Alpha Vantage FX & Commodities	FX + commodities	Medium	Free but limited

6. News Signals (For Macro + Stock + Sector)

For sentiment + breaking news detection.

API	Coverage	Ease	Cost
NewsAPI	Global business news	Very easy	\$
Google News RSS	Free	Very easy	Free

Technology Stack Planned:

1. Data Acquisition (Fetching Market Data): Python
2. Signal Detection Engine: Python + Pandas/Numpy for signal logic
3. Queue Processor / Background Jobs: Celery + Redis (Python)
4. Blog Content Generation (AI): OpenAI GPT API (3.5 or 4o-mini)
5. Template Mapping + Draft Generation: Jinja2 Templates + Python
6. Review → Approve → Publish Workflow: Django REST Framework + Django Admin